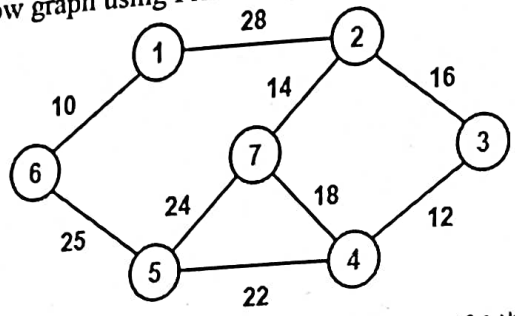
4/9/2022

6/6

Assignment-1

4. Explain the different steps used to find the minimum cost spanning tree for the below graph using Prim's algorithm.(6 Marks)

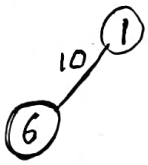


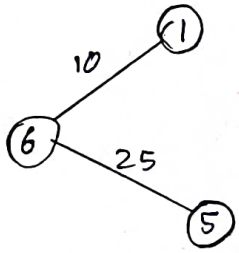
initially,

{0, INF, INF, INF, INF, INF, INF}
= key set {} = mst set

From 1; (1,2) = 28 < INF
(1,6) = 10 < INF
minimum (1,6) add mstset

Key set = {0, 28, INF, INF, INF, INF, INF} mst set = {1, 6}



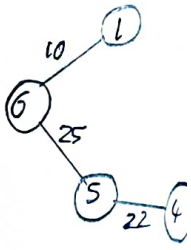


From 1, From 6
(1,2) = 28, (6,5) = 25 < INF
minimum (6,5) = add mst set
Key set = {0, 28, INF, INF, 25, 10, INF}
mstset = {1, 6, 5}

From 1
(1,2) = 28

From 6
now new edges
so no need further

From 5
(5,7) = 24 < INF
(5,4) = 22 < INF
update keyset

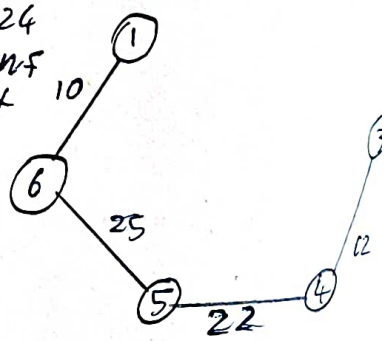


minimum = (5,4) add mstset
Keyset = {0, 28, INF, 22, 25, 10, 24}
mstset = {1, 6, 5, 4}

From 1
(1,2) = 28

From 5
(5,7) = 24

From 4
(4,7) = 18 < 24
(4,3) = 12 < INF
update keyset



minimum (4,3) add mstset
key set = {0, 28, 12, 22, 10, 18}
mstset = {1, 6, 5, 4, 3}

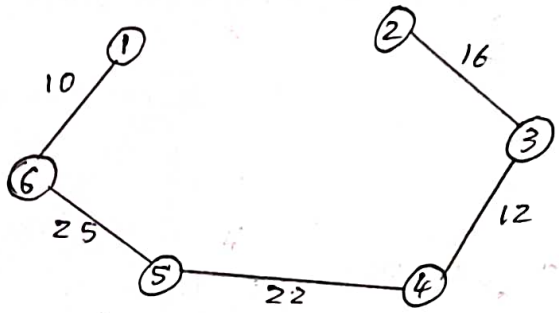
From 1 | From 5 | From 4 | From 3

(1,2)=28 | (5,7)=24 | (4,7)=18 | (3,2)=16 < 28 update key set

Pick (3,2) minimum on mst set

keyset = {0,16,12,22,25,10,18}

mstset = {1,6,5,4,3,2}



From 1
we got minimum edge to 2 so no need to consider further (1,2)=28

From 3
no new edge

From 5
(5,7)=24

From 4
(4,7)=18

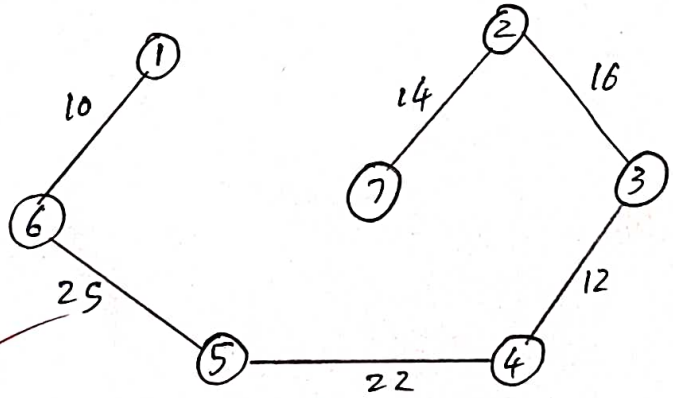
From 2
(2,7)=14 < 18 update key set

Pick (2,7) minimum on mst set

keyset = {0,16,12,22,25,10,14}

mstset = {1,6,5,4,3,2,7}

all vertices are available on mst set so stop the process



Total weight = 99